

Education

New York University, New York, NY

Aug 2024 – May 2026

Master of Science in Computer Engineering (GPA: 3.9/4.0)

Coursework: Machine Learning, Deep Learning, Big Data, System Design, Computer Architecture, Data Structures

Experience

- CampusMesh** Jan 2026 – Present
AI Software Engineer
– Implemented **machine learning-powered features** via **LLM retrieval pipelines (RAG)**, applying **embedding-based semantic search** to improve contextual accuracy by **30%** for intelligent assistant workflows.
– Built and maintained **ML-serving infrastructure** on AWS (**Lambda, API Gateway, S3, CloudFront**) across **4 environments**; wrote clean **Python/Node.js** services backed by **MySQL** and **DynamoDB**.
– Supported testing, debugging, and deployment of production services; participated in code reviews and team knowledge sharing across backend workflows.
- Amazon** May 2025 – Aug 2025
Software Development Engineer Intern
– Contributed to **intelligent system development** by implementing a **RAG pipeline** with an **agentic workflow** that retrieves operational logs and generates context-aware remediation steps.
– Built scalable **ML infrastructure components** on **AWS ECS Fargate, SQS, SNS** with **AWS CDK**; supported testing, debugging, and deployment across auto-scaling distributed clusters.
– Engineered backend services in **Java** with **unit and integration tests (Mockito)**; collaborated with senior engineers through code reviews and design discussions.
- New York University (Novel AI Technologies, Inc.)** Aug 2024 – Present
AI Software Engineer
– Implemented **machine learning models** for an **NSF-funded (\$100K)** health platform, including **Isolation Forest** and **LSTM** time-series anomaly detection on real-time wearable telemetry.
– Built end-to-end **deep learning pipelines (YOLOv8, EfficientNet)** for image classification; deployed models on cloud-hosted inference endpoints with monitoring dashboards.
– Engineered **Python ETL pipelines** into **PostgreSQL**; processed **250K+** records and powered **AWS QuickSight** dashboards for model output monitoring and analytics.
- PricewaterhouseCoopers (PwC)** Aug 2021 – Aug 2024
Senior Software Engineer
– Developed production software in **Java** and **Python** on **Microsoft Azure**, supporting **ML inference pipelines** for document processing and analytics integrations.
– Built an **NLP deep learning pipeline** using **LayoutLM transformer** and **Azure Form Recognizer** to automate structured data extraction, reducing processing time by **75%**.
– Optimized data ingestion with **parallel processing** on **100K+** records; led code reviews and mentored engineers on testing and software design, reducing production bugs by **25%**.

Projects

- RoBERTa Fine-Tuning with Custom LoRA** | *Python, PyTorch, Hugging Face, scikit-learn, AGNews*
– Implemented a **parameter-efficient fine-tuning (PEFT)** pipeline with custom **LoRA** on **RoBERTa-base**, applying **ML optimization techniques** (AdamW, linear decay) under strict memory constraints.
– Achieved **83.45%** classification accuracy while reducing trainable parameters to **934K** via frozen pretrained weights and rank-decomposition matrices.
- InviLite (ML Pipelines & Intelligent Systems)** | *Python, AWS, SageMaker, Bedrock, FAISS, scikit-learn*
– Architected cloud-native **ML data pipelines** (Kinesis, Lambda, S3, Glue, DynamoDB) ingesting structured and unstructured data into governed training and inference workflows.
– Built a **RAG system (SVM + FAISS + Bedrock LLM)**, lifting classification accuracy from **56% to 92%** through feature engineering and model retraining.
- High-Efficiency ResNet Image Classifier** | *Python, PyTorch, TensorFlow, TensorBoard, CIFAR-10*
– Designed a **ResNet** architecture for **CIFAR-10** under a **5M parameter** limit; applied augmentation and the **Lookahead optimizer** with **TensorBoard** for convergence analysis.

Skills

- Programming Languages:** Python, Java, C++, JavaScript (ES6), TypeScript, SQL
- AI/ML:** PyTorch, TensorFlow, scikit-learn, Hugging Face, Transformers, Deep Learning, LLM Fine-tuning, RAG (LangChain, FAISS), NLP, Computer Vision (YOLOv8, EfficientNet), Pandas, NumPy
- Web & Backend:** React, Node.js, Express.js, REST API, Microservices, Software Development
- Databases:** PostgreSQL, MySQL, MongoDB, DynamoDB
- Cloud & DevOps:** AWS (Lambda, SageMaker, Bedrock, ECS, S3, CloudWatch, CDK), Azure (AI Services), Docker, Kubernetes, CI/CD, Git
- Tools:** Jupyter, Linux, Mockito, TensorBoard, Postman, Model Context Protocol (MCP)

Publications

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques (*IEEE*)

An Aquila-optimized SVM Classifier for Diabetes Prediction (*IEEE*)

Certifications

AWS Cloud Practitioner

Stanford Machine Learning (Coursera)

Azure Data Fundamentals